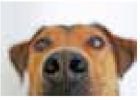
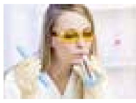


Sniffing Out The Contagion
Researchers have successfully trained six detection dogs to sniff out SARS-CoV-2 in our sweat. **http://dgit.in/jan21-11**



Scratch-and-sniff Test
Researchers are developing an at-home scratch-and-sniff test for COVID-19 **http://dgit.in/jan21-12**

that cause the common cold, but does develop long lasting immunity against the more harmful strains, such as the viruses responsible for SARS and MERS. Over time, the amount of antibodies in the blood after an infection reduces after a person is diagnosed with COVID-19, the same as SARS and MERS. However, this is no indication that the actual immunity of the individual has reduced. Antibody testing does not reveal any data on the other components of the immune system, including memory cells and T cells. In all the known cases of reinfection, the tests looked for viral RNA load, which can continue to be introduced into the blood even after the body has fought of the infection. While reinfection can take place, it is typically mild or asymptomatic. This is consistent with the known cases of reinfections, where there is only one exception. This patient had a more adverse reaction during the second infection as compared to the first. One of the possibilities is that the patient in question had a strong response from the innate cells, providing a first line of defence which sufficiently combated the first infection, but the body did not have enough time to develop a complete immune response, along with the lymphocytes that produce antibodies and remember the virus. Researchers from the University of Manchester stress on the importance of using additional testing such as viral DNA sequencing and capturing data on T cells or memory cells in cases of reinfection, to better understand the disease.

Researchers from the University of Michigan are embarking on a widespread study of reinfections. The study hopes to answer some of the fundamental questions of reinfections, such as if a reinfection is possible? What happens during a reinfection? Does the person get sick or not get sick? Does the body transmit the virus or not during a reinfection? During the course of the study, the researchers will be looking at correlates of protection including T cells, and not just at the antibodies.

IMPROVEMENTS IN PROTECTION

Researchers from MIT have significantly improved the efficacy of face masks in protecting from the coronavirus by introducing a heated element that neutralises the virus. The heated copper mesh slows down and inactivates any virus as the person wearing the mask breathes in and out. The masks are designed to be used by healthcare professionals, or people moving through crowds, such as mass transit systems. In March, the researchers realised that there were no masks that actually killed the virus, and that all the masks only captured it. Instead of filtration, the mask uses an approach known as thermal inactivation, where heat is used to destroy the virus. After investigating the suitable temperature range to deactivate the virus, the researchers found that at 90 degrees celsius, the viral load could be reduced from between a thousand to a million times. Bearing in mind the comfort of the wearer, the air is subsequently cooled. As the masks kill the virus, they do not need to be decontaminated or thrown away. Another advantage of the mask is that not only is the wearer breathing in medically sterile air, they are also breathing out clean air. One of the researchers on the project, Michael Strano says, “This is a completely new mask concept in that it doesn't primarily block the virus. It actually lets the virus go through the mask, but slows and inactivates it”.

Credit: MIT

Heated face masks to neutralise the cornavirus

Credit: University of Wisconsin-Madison

A “badger” mask fitter

One of the problems with masks is achieving the perfect fit, which can be difficult especially in most of the models that do not have the thin aluminum strip to wrap around the nose. Even then, there are gaps left along the cheekbones and the chin. To solve this problem, engineers from the University of Wisconsin-Madison have developed a number of mask fitters, that sit on top of the mask and ensure a tight seal. Researchers arrived at the design of the fitter by iteratively designing increasingly effective solutions. The fitters are made up of low cost plastic and elastic components, and are designed to be used by anyone in public facing jobs. The fitter can also be used on top of any underlying mask. During the research, the scientists found a rather impractical method of making sure there was a tight seal, that was more effective than the fitter – taping the mask to the face!